

M/M/c Erlang C Stochastic Wait-Time Estimation

Parameterized multi-server queue modeling with mathematically bounded p50 / p90 percentiles

1. Poisson Arrivals

Rate: λ vehicles/hr

- Time-of-day demand
- Traffic density proxy
- EV fleet adoption rate

Input Parameter λ

2. Multi-Server Ports

Capacity: c ports, μ min

- Intensity: $\rho = \lambda / (c \times \mu)$
- Stability check: $\rho < 1.0$
- Exponential service time

Erlang C Delay Prob $C(c, \rho)$

3. Percentile Waits

p50 & p90 Distribution

- Median wait: p50 (mins)
- Tail wait: p90 (mins)
- Conditional exponential

Output Wait Estimates

4. Invariant Mesh

p50 $\leq$ p90 Always

- 728 test meshes
- Zero violations
- Overload bounded

✓ 100% Correct

Mathematical Verification: When Erlang C Delay Probability < 0.5 , over 50% of arrivals experience zero wait ($p50 = 0$), while tail queueing is captured by $p90 > 0$.